MOHAB ATEF EL-REFAAI

CONTACT

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Location: Egypt, Al-Dakahlia

SKILLS

Programming & Embedded

- C/C++
- Python
- Arduino
- ESP32
- ATmega32
- Embedded Systems
- I2C / UART

Engineering

- Sensors & Instrumentation
- Control Systems
- Robotics
- Circuit Prototyping
- Mechanical Design

Software

- ImageJ
- LTspice
- Arduino IDE

LANGUAGES

Arabic (Fluent)

English (Fluent)

Japanese (Basics)

PROFILE

Third-year Mechatronics Engineering student with a strong interest in embedded systems, automation, robotics, and intelligent control systems. Experienced in developing Arduino-based projects and applying engineering principles to solve practical problems. Actively involved in technical and professional student organizations, continuously developing both technical expertise and teamwork skills while seeking opportunities to gain industry experience through internships.

EDUCATION

Egypt-Japan University of Science and Technology Sep 2024 – Jun 2028

B.Sc., Mechatronics, Robotics, and Automation Engineering
Alexandria, Egypt

Activities and societies: Member of Robotics Club, Xcellence Club, and IEEE Student Branch. Engaged in technical workshops, engineering activities, teamwork, and professional development initiatives.

PROJECTS

IoT Solar Monitor April 2026 – EJUST

- Developed a microcontroller-based hardware testing rig to evaluate solar panel efficiency across varying thermal and humidity conditions.
- Managed analog signal acquisition by routing voltage and current sensor outputs through a dedicated I2C-based 16-bit analog-to-digital converter.
- Resolved hardware-level communication requirements, including configuring I2C data lines (SDA/SCL) and managing logic-level constraints for 5V sensors interfacing with a 3.3V microcontroller.

Autonomous Robotic Vehicle June 2026 – Personal Project

- Programmed C/C++ control logic on a microcontroller to process real-time distance data from an ultrasonic sensor, driving DC motors via a motor driver for dynamic obstacle avoidance.

Automated Gas Detection & Alert System August 2026 – Personal Project

- Engineered a microcontroller-based safety monitoring system by interfacing an Arduino with an atmospheric gas sensor to detect hazardous leaks.
- Developed C/C++ threshold logic to continuously sample environmental data, triggering a localized piezo buzzer and red LED warning when gas concentrations exceed safe limits.
- Implemented continuous visual state-monitoring using a green LED to indicate safe baseline conditions and active system functionality.

EXTRACURRICULARS & LEADERSHIP

- **AEE:** Vice Head of the Smart Buildings, BMS and Automation Track.
- **SWUG (SolidWorks User Group):** HR Member.
- **E-JUST Microscopy Club:** Member.
- **Xcellence Club:** Member.